Kanav Dutta

+1 (587) 545-5631 kdutta1@ualberta.ca linkedin.com/in/kanavdutta github.com/kanavdutta

Summary

Computing Science student at the University of Alberta developing VR research software for memory studies. Hands-on experience with Python, Java, C#, local LLMs, embeddings, and agentic systems, alongside production software internships. Research interests include immersive VR, AI agents, and quantum information science.

Education

B.Sc. in Computing Science Specialization Edmonton, AB
University of Alberta *2022 - 2027*
Areas of Study: Software Engineering, Web Applications & Architecture, Databases, Algorithms, Cloud Computing.

Technical Skills

Languages	Python, Java, C#, TypeScript, JavaScript, Go, SQL
Research Software	Unity, SteamVR, C#, 360-degree video, VR development
Backend	Node.js, REST APIs, schema validation, async queues
AI & GenAI Tooling	Claude, Ollama, local embeddings/RAG, agentic tool-calling systems
DevOps & Tools	Azure, Git, GitHub, Docker, CI/CD
Databases	PostgreSQL, SQLite, MongoDB

Experience

Student Research Developer Aug 2026 - Present
Memory for Events Lab, University of Alberta *Edmonton, AB*

- Modernizing and extending a Unity VR application for autobiographical memory research, upgrading from Unity 2020.3 to a newer engine version while preserving experimental behavior.
- Developing C# scripts to present 2D/3D 360-degree video stimuli through SteamVR headsets and collect presence and memory ratings for a stereoscopic-depth memory study.
- Collaborating with the principal investigator and lab members to implement repeatable VR experiments.

Software Engineering Intern Sep 2022 - Sep 2023
Adobe Inc. *Edmonton, AB*

- Collaborated with development, QA, and project management colleagues through Agile standups and sprint cycles.
- Built and shipped full-stack TypeScript and JavaScript features, including React components, REST API integrations, and backend logic, supported by automated unit tests and peer code review.
- Diagnosed a data-fetching bug affecting state across 3 components; improved page performance by 15% and eliminated recurring user-facing errors.

Software Development Intern Sep 2021 - Mar 2022
Fore Solutions Pvt. Ltd. *New Delhi, India*

- Delivered full-stack features within sprint deadlines, taking ownership from design through testing and deployment.
- Re-architected a data processing pipeline, reducing turnaround time by 57% while maintaining output integrity, and documented the change for ongoing team reference.

Projects

Local AI Agent Toolkit *Python, Ollama, SQLite, Local Embeddings, Agentic AI*

- Built a suite of local AI tools in Python using Ollama, including a general-purpose assistant.
- Integrated SQLite-backed memory and local embeddings to support persistent assistant context and retrieval.
- Implemented a tool-calling agent loop that plans and executes multi-step tasks.

AssistAI *TypeScript, React Native, Node.js, REST APIs, PostgreSQL, Azure, JWT Auth*

- Built a full-stack application with REST API endpoints, JWT authentication, and structured request/response validation.
- Deployed backend services to Azure through a CI/CD pipeline that ran the full unit test suite on every commit.

Cypher Event Lottery System *Java, Android, Firebase | CMPUT 301*

- Built a Java Android application for lottery-based registration for high-demand community events.
- Implemented organizer workflows for creating and managing events, waiting lists, and invitations.
- Developed entrant workflows for scanning event QR codes, joining waiting lists, and responding to invitations.

References

Peggy | Research Head, Memory for Events Lab | pstjacqu@ualberta.ca

Owen Randall | Professor, University of Alberta | davidowe@ualberta.ca

Eric Nielsen | Director, UASU | eric.nielsen@su.ualberta.ca